Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M.Tech. in Artificial Intelligence and Machine Learning.

I Semester 2025-26

Course Number	AIMLZC416
Course Name	Mathematical Foundations for Machine Learning
Nature of Exam	Open Book
Weightage for grading	30%
Duration	150 minutes
Date of Exam	//2025 (FN: 09:00 - 11:00)

# Pages	3
# Questions	7

Instructions

1. All questions are compulsory
2. Marks are indicated against each question

Q1. Two particles on a 2D plane start at the origin at time $t = 0$. One proceeds with velocity u along the direction $\boldsymbol{p}$ and the other proceeds along the direction $\boldsymbol{q}$ with velocity$v$. Express the rate of change of the distance between the two particles at time t in terms of $\boldsymbol{p}, \boldsymbol{q}, u, v$. If the angle between the two unit vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ is θ, find the magnitude of the rate of change of the distance between the two particles at time t in terms of u, v and θ. (6 marks)

Q2. It is known that the characteristic polynomial $p_A(\lambda)$ of a $n \times n$ matrix A can be written such that the smallest power of λ in it is 3. Can you draw any conclusion about the determinant of A from this information? If A is subjected to Gaussian elimination resulting in its REF form, can you obtain a lower bound on the number of zero entries in the REF form of A? If the rank of A can be stated to be $p \leq rank(A) \leq q$, can you obtain p and q? Give suitable justification for all your answers to the given questions. (6 marks)

Q3. Consider the following quadratic optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + 2x_1x_2 + 2x_2^2 + 3x_1 + 2x_2 \\ \text{subject to} \quad & 7x_1 + 9x_2 \leq 1 \\ & x_1 + 5x_2 \leq 3 \end{aligned}$$

(1) Rewrite the optimization problem in the following quadratic programming form discussed in MFML course:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & \tfrac{1}{2}\mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b}, \end{aligned}$$

where Q is a symmetric matrix. Mention clearly $\mathbf{Q} \in \mathbb{R}^{2\times 2}, \mathbf{A} \in \mathbb{R}^{2\times 2}$ and $\mathbf{b}, \mathbf{c} \in \mathbb{R}^2$ in your solution.

(2) Derive the Lagrangian of the reformulated optimization problem. Derive the minimum of Lagrangian with respect to $\mathbf{x}$.

(3) Derive the dual function $D(\lambda_1, \lambda_2)$ where λ_1, λ_2 are the Lagrange multipliers.

(4) Derive the dual optimization problem.

(1+2+2+1 marks)

Q4.(a) Consider a binary classification problem where the two possible classes are +1 and -1. You are given a linear classifier defined as $f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$ where $\mathbf{w} = \begin{bmatrix}1 & 1\end{bmatrix}^T$ and $b = -3$. Calculate hinge loss for the binary classification problem where the prediction is given by $f(\mathbf{x})$ for the following four data points: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ and $\mathbf{x}_4$.

i) $\mathbf{x}_1 = [3 \;\; \frac{1}{2}]^T$

ii) $\mathbf{x}_2 = [4 \;\; 2]^T$

iii) $\mathbf{x}_3 = [0 \;\; 0]^T$

iv) $\mathbf{x}_4 = [1 \;\; 1]^T$

Assume that the true class of $\mathbf{x}_1, \mathbf{x}_2$ is '+1'. Also assume that the true class of $\mathbf{x}_3, \mathbf{x}_4$ is '-1'.

(2 marks)

(b) Let $\mathbf{x} \in \mathbb{R}^n$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$. Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be constants. Consider the following optimization problem

$$\min_{\mathbf{x}} - \sum_{i=1}^{n} log(\alpha_i + x_i)$$

$$\text{subject to } \sum_{i=1}^{n} x_i = 1$$

$$x_i \geq 0 \quad \text{for all } i$$

i) Derive the Lagrangian of this constrained optimization problem.

ii) Derive the four KKT conditions for the above optimization problem.

(1+3 marks)

Q5. Consider the following dataset.

x	y
1	5
2	7

To estimate $\hat{y}$, assume the linear model $\hat{y} = x\,w$.

(a) Write the Mean Squared Error loss function $L(w)$ for this dataset. (3 Marks)

(b) Perform two iterations of the AdaGrad update

$$w_{i+1} = w_i - \frac{\alpha}{\sqrt{A_i + \varepsilon}} \frac{\partial L}{\partial w}\bigg|_{w=w_i},$$

using the initial values $w_0 = 0$, $A_0 = 0$, learning rate $\alpha = 0.2$, and $\varepsilon = 10^{-8}$. (3 Marks)

Q6. Consider the data given below.

3	2	1	4	7	6
1.2	1.4	1.7	1.8	2.0	2.1

(a) Construct the covariance matrix for the above data. (2 Marks)
(b) Find the first principal component of the covariance matrix. (2 Marks)
(c) Using the first principal component from part (b), compute the reduced one-dimensional representation of the six observations. (2 Marks)

Q7.(a) Let A be a matrix such that its eigen decomposition is the same as singular value decomposition. Prove or disprove that A is symmetric.

(b) Given that the matrix

$$A = \begin{bmatrix} 1 & a & 0 \\ a & 5 & 2 \\ 0 & 2 & b \end{bmatrix}, \quad a, b \in \mathbb{R},$$

has 2 and -1 as its eigenvalues with algebraic multiplicities 2 and 1, respectively, find the values of a and b.

(c) If the eigenvalues of a matrix are real and contained in the disc $|\lambda| < 1$, then prove or disprove that $A^n \to \mathbf{0}$ as $n \to \infty$. (1 + 2 + 1 Marks)